CTI Report

Ghost in the Front End

Anatomy of the Microsoft 365 outage of 12–13 November 2026: what failed, how it spread, why it was not an attack, and what defenders should change.

Author	Issued	Version	Information cut-off	Marking
Douglas Mun with AI assistance	13 Nov 2026	1.2 (full report)	09:10 UTC 13 Nov (17:10 SGT)	TLP:CLEAR



Trigger: a routing change is the leading suspect (KJ-2). Amplifier: health checks (KJ-3). Same traffic as usual, far fewer open gates.

Was it a cyberattack?

No evidence of one.

Everything Microsoft has disclosed describes an operational failure inside Exchange Online. In four successive updates it has said it is not aware of any malicious activity, and a hacktivist claim circulating online does not hold up. We judge a malicious cause **very unlikely**, with Moderate confidence (KJ-4; page 11).

16 h 53 min	138,058	20	94 min
of degraded service at our cut-off, from 16:17 UTC	peak Downdetector reports for Outlook, near 18:00 UTC	Microsoft 365 services on Microsoft's affected list	from the first admin-centre notice to Microsoft's first public post

Six questions, short answers	
Was it a cyberattack?	No evidence of one. Microsoft has four times said it knows of no malicious activity, and the failure pattern is operational (KJ-4).
What broke?	Capacity in Exchange Online's client-access tier. Everything that reads Exchange data failed; Teams calls and SharePoint files kept working (KJ-1).
Why did it last so long?	Health checks pulled struggling servers and pushed their load onto the rest; the spiral kept going after the suspected trigger was reverted (KJ-2, KJ-3).
What should users do?	Nothing to their devices: no reinstalls, no password resets. Use phone or chat for urgent matters and ignore "log in again" links.
What should organisations do?	Record what failed today; after recovery, check security tools for gaps and re-run automations without duplicates (page 14).
When will it be fixed?	Not yet. At 09:10 UTC Microsoft saw early signs of stabilisation, but capacity was below normal. Delayed mail is held for 48 hours, not bounced.

Disclaimer

Preliminary. Microsoft has published no root cause; it has committed to a post-incident report (PIR) once the incident is mitigated. These judgements rest mainly on Microsoft's service-health updates, read from public mirrors of its API and from @MSFT365Status, and will be re-graded when the PIR appears. Times are UTC, with Singapore time (SGT) in brackets.

Changes since version 1.0 (10:00 SGT flash)

- Extended from 02:00 to 09:10 UTC: the 01:45 recovery failed; Microsoft named a "traffic influx", added capacity, patched a routing component and saw early stabilisation.
- KJ-1 restated as confined to Exchange Online; KJ-2 lowered to Low confidence after servers kept failing past the network revert.
- Corrected the Teams incident window (14:45–15:53 UTC) and the Home Affairs Minister's statement.
- Expanded into the full report: ACH, Dark Storm Team claim, consumer, sector and AI impact, precedents and tiered response, partly drawing on Data is CITY Media.

Keys

Confidence in a judgement

High Moderate Low

Status of a fact (page 16)

Confirmed Reported Assessed Open

Attack indicator, ACH (page 11)

Not seen Open

Confirmed: in Microsoft's own record. Reported: from press or mirrors only. Assessed: our judgement. Likelihood follows ICD 203: very unlikely 5–20%, unlikely 20–45%, likely 55–80%, very likely 80–95%.

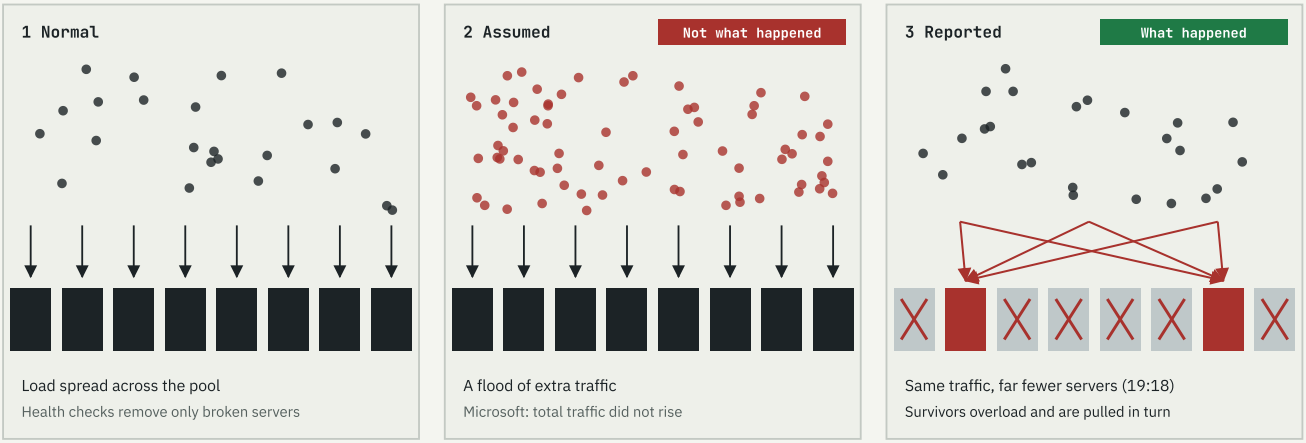
Seventeen hours in, early stabilisation; capacity is not yet back to normal

From 16:17 UTC on 12 November (00:17 SGT, 13 November), Exchange Online's servers have lost capacity faster than Microsoft can restore it. Outlook is down, and with it every Microsoft 365 feature that reads Exchange data, for organisations and for consumers on Outlook.com. Microsoft narrowed the cause from load, to health checks, to an overwhelmed secure-connection process, and at 01:06 UTC began reverting a network change that coincided with the onset. Restarts looked promising at 01:45 UTC but did not hold. By 07:00 it reported only "marginal success" and was adding capacity; at 07:29 a patch to an upstream authentication/routing component began showing positive results; at 09:10 it saw early stabilisation and extended the email timeout to 48 hours to protect delayed mail.

What Capacity collapse inside Exchange Online, tracked as EX1550543 (Exchange Online) and MO1550821 (Microsoft 365 suite).	**When** Impact from 16:17 UTC 12 Nov (00:17 SGT 13 Nov); still degraded at 09:10 UTC (17:10 SGT). A separate Teams incident ran 14:45–15:53 UTC.	**Who** Mailbox access over Outlook on the web, MAPI, ActiveSync and REST, plus Exchange-dependent features of 19 other services. The public too: Outlook.com, OneDrive, OneNote, To Do, Teams and Microsoft 365 for the web.
Where Americas first, then any region. Singapore was hit from 00:17 SGT, and the outage has run through the working day.	**Why** Not established. A routing change is the leading suspect (KJ-2); health checks amplified it (KJ-3). Microsoft promises a root cause after recovery.	**Scale** Downdetector reports for Outlook peaked at 138,058 near 18:00 UTC (02:00 SGT). Reports count complaints, not affected users.

What failed: fewer servers, not more traffic

The instinctive reading of a global outage is a flood: too many requests, or an attack. Microsoft ruled that out at 19:18 UTC. Total traffic had not risen; a large number of servers had become unavailable, and their share of the load landed on the rest. The figure shows the difference.



Illustrative. Microsoft has not published server counts.

A Singapore analogy: the fare gantries at peak hour

A sensor marks one MRT fare gantry faulty and shuts it. Its commuters queue at the gantries beside it, which jam under the extra load and are shut in turn. The crowd never grew; the number of open gantries shrank, and every closure made the next one more likely. Microsoft's front-end servers did the same, with automated health checks as the sensors. Reopening a gantry does not help if the queue knocks it straight back out, which is why Microsoft's restarts kept failing.

Key judgements: four, all provisional until Microsoft's post-incident report

KJ-1 **The failure is confined to Exchange Online's serving path; shared Microsoft 365 sign-in is not down.** Very likely **High**

Microsoft's list draws the line precisely: Teams calls, meetings and chats, and SharePoint files, keep working, so shared sign-in works, while everything that reads Exchange data fails. Entra ID has never been listed. The 07:29 fix targets an "upstream authentication/routing configuration component": something in front of the "backend servers" of 05:46 that authenticates and routes requests. That is the role Exchange gives its client-access front end (CAFE), so the front-end placement is likely, though Microsoft has not named it.

Gap: a PIR naming a shared identity service or Azure Front Door as the failed component would overturn this judgement (Q-7).

KJ-2 **A change to how traffic is routed into Exchange probably started the failure.** Likely **Low**

Microsoft began reverting a network configuration change that correlates with the onset at 01:06, after asking at 18:57 whether routing had shifted. Since then it has described "sustained traffic management imbalances" (03:13) and a "traffic influx" (05:46), and at 07:29 the first fix to show results was a patch to a routing configuration component. The revert alone did not end the outage, which fits a cascade that outlives its trigger. By 09:10 Microsoft reported early signs of stabilisation without crediting the patch, so the test is not yet passed.

Gap: if the 07:29 patch holds as it expands, raise confidence to Moderate. The PIR must show whether the routing component was the trigger or only the cure (Q-1, Q-4, Q-7).

KJ-3 **Health checks were both a true alarm and an accelerant; a health-check bug alone is unlikely to be the root cause.** Unlikely **Moderate**

Changing the check returned a subset of servers to service (20:56), so the checks were removing capacity. But returned servers failed again once traffic reached them (23:59), after high CPU (19:57) and an overwhelmed core process (22:00), later named as the process that sets up secure connections. A server that fails under load is not healthy.

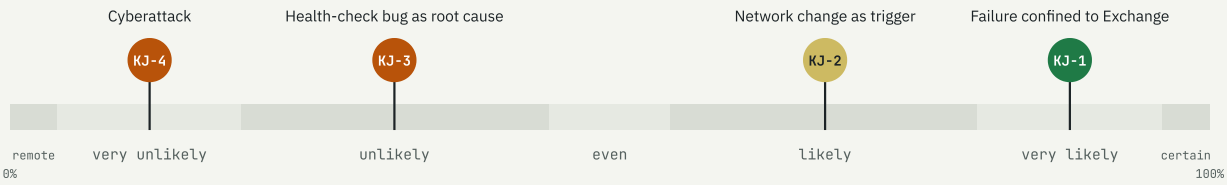
Gap: Microsoft has not said whether the health-check component shares the secure-connection process or its resources. If it does, the two explanations describe one fault (Q-3).

KJ-4 **A cyberattack is very unlikely.** Very unlikely **Moderate**

In four updates from 05:46 to 09:10 Microsoft stated it is not aware of any malicious activity causing the incident, and it told Australia's Home Affairs Minister it did not believe a cyberattack was the cause. Its fixes remain operational: restarts, rollbacks, load management and software patches. The Dark Storm Team claim circulating online offers no evidence of cause (page 3). No source links the outage to CVE-2026-124194.

Gap: confidence stays Moderate because Microsoft is the only primary source and has not explained the "traffic influx" (05:46) or the traffic type it examined at 18:27. Evidence that either was hostile would reverse this judgement (Q-5).

Where our four judgements sit on the likelihood scale



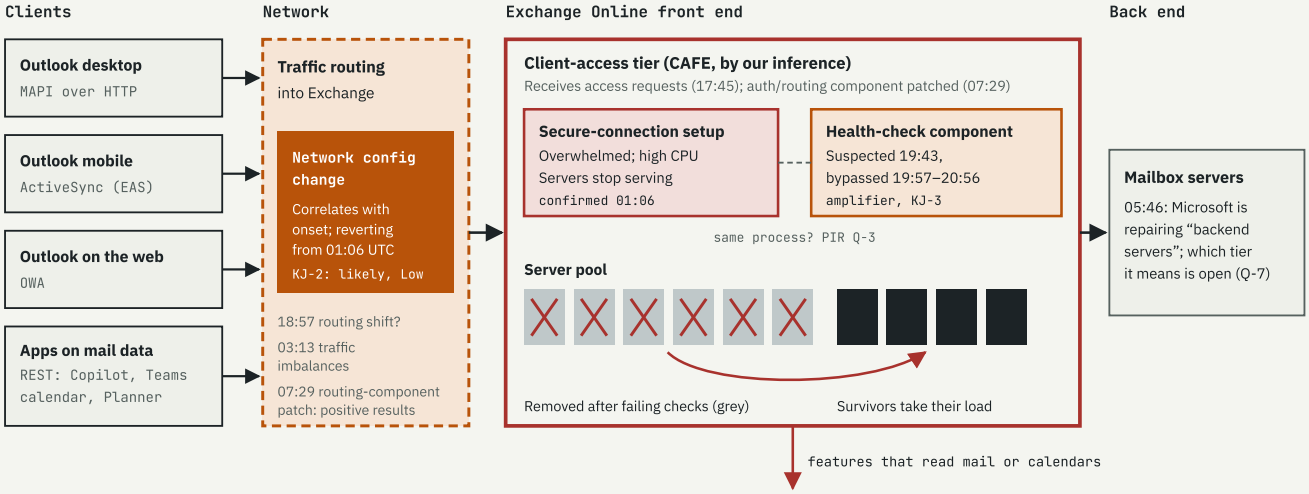
Marker colour shows confidence: green High, orange Moderate, yellow Low.

Five signals that would change these judgements

Signal	Where to look	What it would mean
Capacity returns to normal and the incidents close	Service health; your own mail flow	Re-grade the judgements; if the 07:29 patch is credited, raise KJ-2 to Moderate
Microsoft explains the "traffic influx"	Service health or the PIR	Settles KJ-2 and the one open attack indicator (KJ-4)
The PIR names the network change and the fix that held	Post-incident report	Raises or retires KJ-2; answers Q-1 and Q-4
A security advisory tied to the incident	MSRC; Microsoft security blog	Would reopen KJ-4 at once
Hacktivist claims or outage-themed phishing reaching constituents	CERT reporting channels	Escalate A-4 from precaution to advisory

Where it broke: Exchange's front door, not Microsoft 365's

Every Outlook client, and every Microsoft 365 feature that reads mail, calendars or contacts, enters Exchange Online through its client-access tier: servers that accept the connection, set up encryption, work out whose mailbox a request belongs to, and route it to the mailbox server behind them. Exchange calls this tier CAFE (Client Access Front End). Microsoft has not used the name, but every symptom it reported sits in that tier, and its 07:29 fix targeted an "upstream authentication/routing configuration component" in front of the "backend servers".



Failed where they touch Exchange (Microsoft's list, 07:00 UTC)

- Exchange Online, all clients
- Teams calendars and events
- Copilot over work data
- Planner and To Do
- SharePoint features using mail
- Microsoft 365 apps: recent files
- Microsoft 365 Education
- Defender for Office 365:
- Safe Links, quarantine
- Defender XDR portal
- Purview: DLP, labels, audit
- Intune: enrolment, check-in
- Graph API for mail data
- Fabric and Power BI labels
- Microsoft 365 admin centre
- Microsoft 365 Apps admin
- Universal Print
- Windows 365 (Customer Key)
- Power Apps mobile sign-in

`Defender, Purview and Intune are on this list: mail-security coverage has gaps (A-3, A-6, A-7).`
`20 services listed; each fails only in its Exchange-dependent features.`

Kept working

Teams calls, meetings, chats
SharePoint and OneDrive files
Anything not reading Exchange data

`Microsoft, 05:46 UTC: non-Exchange elements remain functional`

Never listed

Entra ID sign-in. A shared identity or front-door fault would have broken Teams chat too.

Also hit: consumer services on personal Microsoft accounts (separate status entry)

Outlook.com, OneDrive, OneNote, To Do, Teams and Microsoft 365 for the web. Outlook.com mailboxes run on the same Exchange Online infrastructure, so the same fault reaches the public. Microsoft describes the impact only as "several actions" failing; file access itself is not confirmed down.

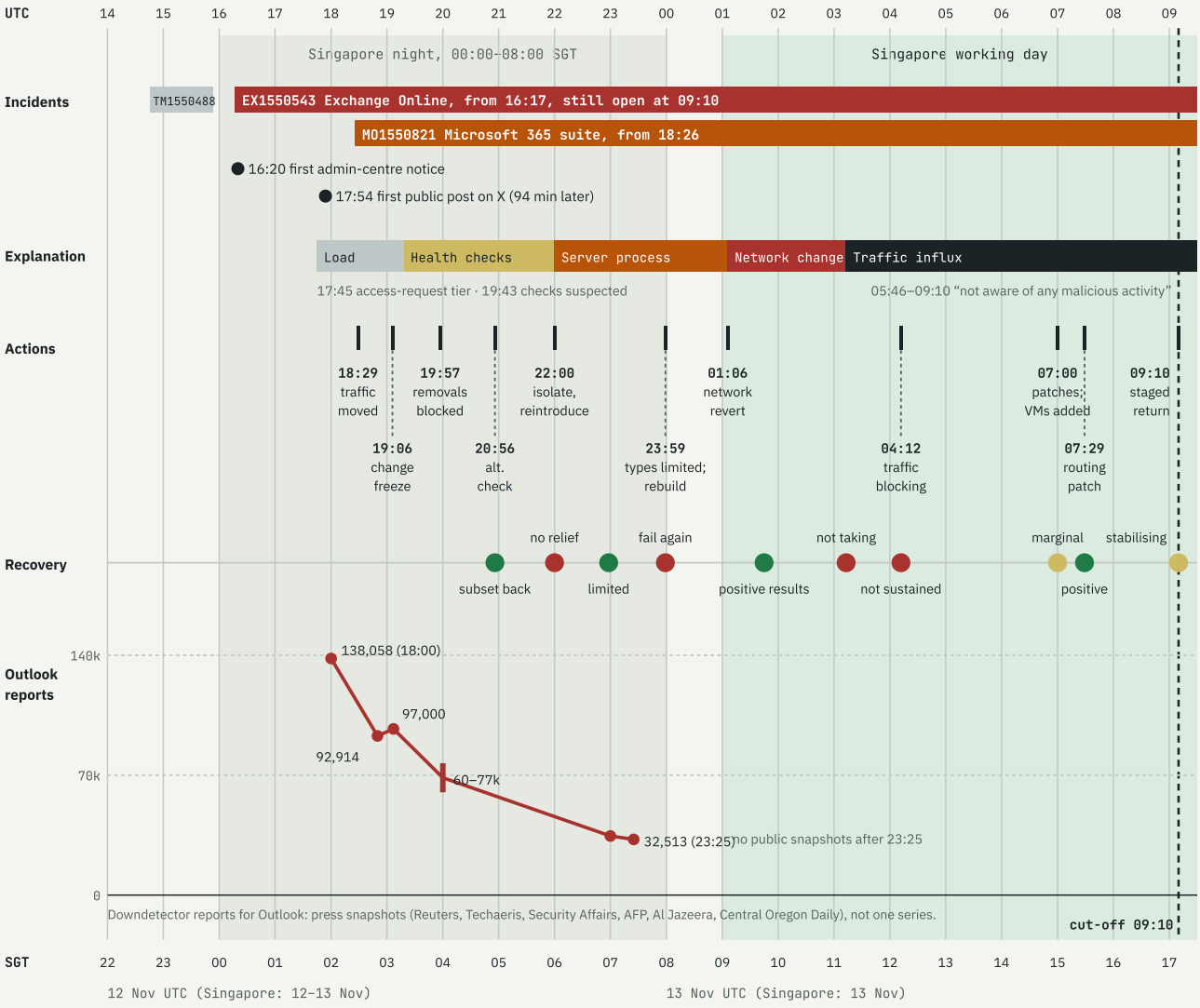
Solid outlines: stated by Microsoft. Dashed outline and "by our inference": our assessment. The server pool is illustrative; Microsoft has not given numbers.

Reading the next Microsoft 365 outage from its symptoms

What users report (our heuristic)	Where the fault most likely sits	This outage
Outlook and mail-based features fail; Teams chat and SharePoint files work	Exchange Online, front end or back end	Match
Sign-in fails across every service	Entra ID, the shared identity layer	No
Every service slow or unreachable, files included	A shared network edge such as Azure Front Door	No

This outage matched the first row from start to finish. Azure's own status page showed no platform-wide incident, and Entra ID was never listed, which is why we place the fault inside Exchange Online and not in Microsoft 365's shared foundations (KJ-1).

Timeline of events: seventeen hours from “elevated load” to “early signs of stabilization”



Singapore slept through the peak; the stalled recovery reached its working day

The outage began at 00:17 SGT and complaints peaked near 02:00 SGT, outside business hours. The working day opened on what looked like a recovery: Microsoft reported positive results at 09:45 SGT. By 11:13 SGT it said its changes were not taking, at 15:29 SGT a routing-layer patch began to help, and at 17:10 SGT Microsoft saw early signs of stabilisation. Expect helpdesk and SOC load through the afternoon, not just after the overnight peak.

Complaint volume measured attention, not recovery

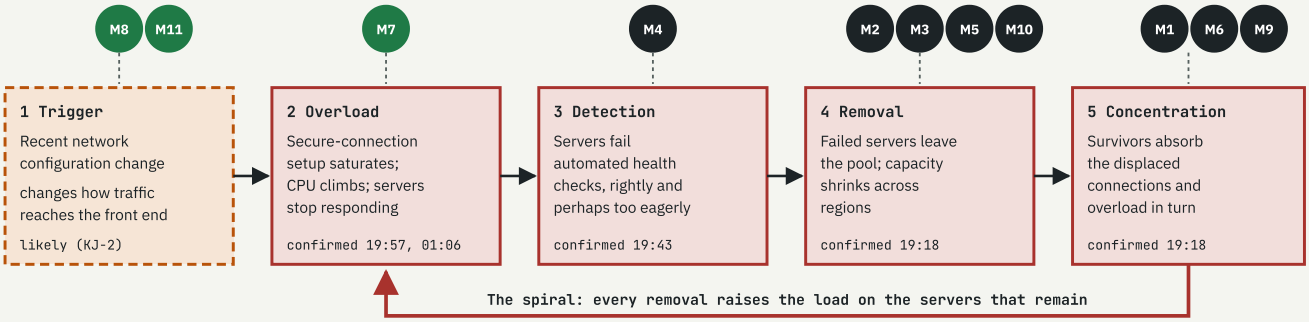
Outlook reports fell by three quarters between 18:00 UTC and the US evening (about 23:00 UTC), from 138,058 to 34,490. Over the same hours Microsoft said its fixes had not brought the expected relief (22:00) and that restored servers were failing again (23:59). The fall tracks users giving up and the US workday ending. By 23:25 UTC reports were down to 32,513, more than seven hours before Microsoft’s 07:00 update reported only marginal success. Do not read a falling Downdetector curve as recovery; use the provider’s status and your own telemetry.

The admin centre beat Microsoft’s public post by 94 minutes

Microsoft’s first notice on EX1550543 reached tenant admin centres at 16:20 UTC. Its first post on @MSFT365Status came at 17:54 UTC, by which time Downdetector reports were near their peak. Organisations watching only public channels lost an hour and a half of warning. Admin-centre alerts, routed to on-call staff, would have given constituents that time back (A-5).

The failure chain: a capacity spiral that health checks accelerated

Each server the health checks removed pushed its connections onto the survivors, which then overloaded and failed their own checks. That loop explains why the outage outlasted every individual fix. The figure maps eleven of Microsoft's recovery measures onto the stage each one attacked.



Cut the cause: M7, M8, M11 (23:59–07:29)
M11, a routing-layer patch, is the first to show positive results (07:29).

Slow the spiral: M1–M6, M9, M10 (18:19–07:00)
Restarts, check changes and traffic moves buy capacity back, but the servers fail again while the trigger is still in place.

Hence each fix gave partial recovery that did not hold (22:00 to 07:00).

Microsoft fought the symptoms for nearly eight hours before it touched a candidate cause

ID	UTC	Microsoft's action	Loop stage	Reported effect
M1	18:29–18:57	Moved traffic off US infrastructure, then redistributed Outlook traffic worldwide	5 Concentration	Added latency; no relief reported
M2	18:19 onward	Restarted affected servers, later all of them (19:43)	4 Removal	Servers returned, then failed again
M3	19:57	Blocked further servers from being removed	4 Removal	Intended to stop the pool shrinking; capacity not restored
M4	19:57–20:56	Bypassed the health-check component, then adopted an alternative check	3 Detection	A subset of servers returned (20:56)
M5	20:56	Prepared automated detection and restart of unresponsive servers	4 Removal	Intended to keep servers in service; no result reported
M6	22:00–23:59	Isolated infrastructure, reintroduced traffic gradually, then limited traffic by type	5 Concentration	Limited recovery in some regions (22:58)
M7	23:59	Rebuilt a subset of servers on a previous software configuration	2 Overload	Not reported at cut-off
M8	01:06	Began reverting the network configuration change that correlates with onset	1 Trigger	Positive results at 01:45 did not hold: changes "not taking as expected" (03:13)
M9	04:12	Adjusted traffic protocols, redistributed traffic and applied strategic traffic blocking	5 Concentration	Partial recovery where applied, not sustained
M10	07:00	Applied newly developed patches after reboots; began adding virtual-machine capacity	4 Removal	"Marginal success"
M11	07:29	Patched an upstream authentication/routing configuration component on part of the infrastructure	1 Trigger	Positive results for server stability; expansion under way
M12	09:10	Cycled servers through drain, restart, remediation and staged return; extended the email timeout to 48 hours	4 Removal	"Early signs of stabilization"; capacity not yet normal

Two further measures sat outside the loop: the change freeze at 19:06 and a configuration change for the secure-connection process that Microsoft was validating on a subset at 01:06. M7 and M8 were a two-arm test of the trigger. Neither had produced a sustained recovery by 07:00, which fits a cascade that keeps itself going after its trigger is removed. At 07:29 a routing-layer patch (M11) showed positive results, and by 09:10 Microsoft saw early stabilisation as it returned servers in stages (M12). The PIR should say which measure finally worked (Q-4).

Walking the failure chain, stage by stage

Each stage below gives what Microsoft reported, how sure we are, and the lesson for organisations that run their own front ends, load balancers and automated recovery.

1 Trigger

likely, Low confidence

A change to how traffic is routed into Exchange

At 18:57 Microsoft asked whether routing, not demand, had shifted. At 01:06 it began reverting a network configuration change that correlates with the onset. It later described “sustained traffic management imbalances” (03:13), and at 07:29 the first fix to show results was a patch to a routing configuration component.

> **Lesson** Treat routing and load-balancing changes like code: staged rollout, health-gated progression and a rollback you have rehearsed. A change freeze helps only if you can tell which change to undo.

2 Overload

confirmed by Microsoft

Setting up secure connections exhausts the servers

Microsoft reported high CPU (19:57), an overwhelmed core process (22:00), and named it at 01:06 as the process that establishes secure connections. Connection setup is the expensive part of serving a request; anything that multiplies new connections, such as lost connection reuse, rerouting or mass reconnection, multiplies load.

> **Lesson** Watch connection-setup rate and CPU on your termination tier, not only request rate. A spike in handshakes with flat requests is an early warning.

3 Detection

confirmed by Microsoft

Health checks fail, rightly and perhaps too eagerly

Servers failed automated health checks (19:43). Microsoft suspected the component that processes them (19:43), bypassed it on a subset (19:57) and adopted an alternative check that brought servers back (20:56). Those servers then failed under traffic, so the checks were partly right.

> **Lesson** A probe that shares threads or CPU with production fails exactly when production does. Isolate probes, and make them test what you care about.

4 Removal

confirmed by Microsoft

Servers leave the pool faster than they return

Servers became “unexpectedly unavailable” (19:18). Microsoft blocked further removals (19:57), restarted servers in waves, and found that restored servers failed again once traffic reached them (23:59). By 07:00 it was adding virtual-machine capacity, and at 09:10 it was cycling servers through drain, restart, remediation and staged return.

> **Lesson** Cap how much of a pool health management may remove, and fail open when most of a fleet fails at once. AWS added exactly this limit after its October 2025 outage.

5 Concentration

confirmed by Microsoft

Survivors absorb the load and fall in turn

The displaced load strained the remaining servers (19:18). Microsoft moved traffic between regions, isolated infrastructure, limited traffic by type (23:59) and applied “strategic traffic blocking” (04:12), with partial recovery that did not hold.

> **Lesson** Recovery needs load shedding and gradual reintroduction. Clients and AI agents that reconnect all at once can knock restored capacity straight back down; build in backoff and jitter.

The record: twenty-one quotations from Microsoft's engineering updates, and what each one tells us

Microsoft's service-health updates are the only primary evidence. Read in order, they show engineers moving from symptom, to mechanism, to suspect. Each quotation below is from an update in Microsoft 365 Service health (the admin centre) or on @MSFT365Status; the time is the admin-centre update that carried it, or the X post where marked.

Phase 1 17:45–19:18 UTC Microsoft saw load, then realised servers were missing

17:45
01:45 SGT
EX1550543

> *"a portion of Exchange Online infrastructure which receives and processes users' access requests"*

The earliest locator is still the most precise. It describes Exchange's client-access front end, the tier every Outlook protocol passes through. Every later update explains why this tier failed; none moves the fault elsewhere (KJ-1).

18:57
02:57 SGT
EX1550543

> *"a shift in how traffic is being routed across our infrastructure"*

Under three hours in, Microsoft asked whether routing, not demand, had changed. Six hours later it named a network configuration change. In hindsight this was the first pointer to the eventual suspect (KJ-2), and the most underrated line of the night.

19:06
03:06 SGT
@MSFT365Status

> *"We've paused all deployments"*

A change freeze is the reflex of an operator that suspects its own pipeline. Microsoft froze changes 2 h 46 min after its first notice, before it had found any change to blame. That is good practice, and it also tells us where Microsoft's instinct pointed from the start.

19:18
03:18 SGT
@MSFT365Status

> *"a number of servers being unexpectedly unavailable"*

The pivotal reframe: a supply problem, not a demand problem, because total traffic had not risen. It is also the strongest single datum against a volumetric attack, which would show the opposite pattern (KJ-4).

Phase 2 19:43–21:00 UTC The health checks became the suspect, then lost sole blame

19:43
03:43 SGT
EX1550543, MO1550821

> *"a recent change to a component involved in processing those checks"*

The first named suspect was the health-check machinery itself. This update is the origin of the health-check-bug theory, including our own first reading of the outage.

19:57
03:57 SGT
@MSFT365Status

> *"We've not identified any recent change correlated with the start of impact"*

Fourteen minutes later Microsoft withdrew the link to a change. The same update reported high CPU on the affected servers and blocked any further removals, a circuit breaker against cascading ejection. Servers burning CPU are struggling, not idle victims of a faulty probe.

20:56
04:56 SGT
@MSFT365Status

> *"validated an alternative service health check method"*

Changing the measuring stick brought a subset of servers back, which proves the checks were removing capacity. It does not prove those servers were healthy, as the next three updates show (KJ-3).

Phase 3 22:00–00:00 UTC The servers themselves were failing under load

22:00
06:00 SGT
EX1550543, MO1550821

> *“actions taken thus far have not provided the anticipated relief”*

A rare, candid admission. The health-check fix had treated the accelerant, not the fire. In the same update Microsoft named a core process on the servers as overwhelmed and began isolating infrastructure to test recovery step by step.

23:59
07:59 SGT
@MSFT365Status

> *“can become unresponsive again once they receive traffic”*

The most important observation of the night. Servers that look healthy at idle and fail under load have a load-dependent defect in the serving path. That rules out the pure form of the theory that healthy servers were wrongly removed (KJ-3). The same update brought networking and platform experts into the analysis.

00:00
08:00 SGT
MO1550821

> *“services that do not depend on interaction with Exchange Online remain functional”*

Microsoft drew the blast radius for us. Teams, Copilot, Planner, Purview and the rest failed only where they touch Exchange; a shared identity or front-door failure would not respect that boundary (KJ-1). The same update rebuilt a subset of servers on a previous software configuration, opening the first of two rollback arms.

Phase 4 01:06–01:45 UTC A process and a change were named; restarts looked promising

01:06
09:06 SGT
EX1550543; MO1550821 at 01:12

> *“a process on the affected servers responsible for establishing secure connections becomes overwhelmed”*

The bottleneck now has a name. Setting up secure connections, most likely TLS sessions, is CPU-heavy work for a front end, which fits the high CPU reported at 19:57. Microsoft has named neither the protocol nor the exhausted resource; that is a question for the PIR (Q-2).

01:06
09:06 SGT
EX1550543

> *“reverting a recent network configuration change that correlates with the start of impact”*

The suspect from 18:57 returns with a specific change attached, the first change Microsoft says lines up with the onset. “Correlates” is a careful word: the PIR must show that the revert, not the restarts or the rebuild, restored stability (KJ-2, Q-4).

01:45
09:45 SGT
EX1550543; X post 01:59

> *“We’re seeing positive results with our efforts to restart affected infrastructure”*

Microsoft credits restarts, with servers carrying a narrower set of connections, not the revert. With both rollbacks and restarts in play at once, recovery alone cannot tell us which one worked. The optimism lasted under 90 minutes (Phase 5).

Phase 5 03:13–07:29 UTC The recovery stalled, and the problem became the traffic itself

03:13
11:13 SGT
@MSFT365Status, X post time

> *“sustained traffic management imbalances”*

The optimism of 01:45 lasted under 90 minutes: Microsoft said its changes were not taking as expected. Its vocabulary also shifted from servers to traffic. The problem was no longer only that servers fail, but that load would not settle evenly across the ones that remain, which is how a cascade sounds once it has outlived its trigger.

Phase 5, continued 04:12–07:29 UTC No fix held until a routing-layer patch

04:12
12:12 SGT
MO1550821

> *“applying strategic traffic blocking measures” … “these actions aren’t resulting in sustained recovery”*

Blocking traffic to protect servers is load shedding, the standard way out of a self-sustaining overload. It will also be read online as attack mitigation. Microsoft frames it as stability work, and recovery stayed partial wherever it was applied.

05:46
13:46 SGT
MO1550821

> *“pinpoint the underlying cause of the traffic influx”*

At 19:18 Microsoft said total traffic had not risen; now it is hunting an influx. The two fit if the influx is local: connections piling onto the servers that remain, through routing imbalance and clients reconnecting, rather than more traffic arriving overall. Microsoft has not said which, and the same update speaks of repairing “backend servers” without naming the tier (Q-5, Q-7).

05:46
13:46 SGT
MO1550821

> *“we are not aware of any malicious activity that is causing this incident”*

Microsoft’s first statement on malice in the incident record, repeated at 07:00 and 07:29. “Not aware” is careful wording, not a forensic finding, but it comes from the only party with the telemetry and matches every observable (KJ-4). The same update promises a post-incident report with “defense-in-depth repair items”: layered fixes, which fit a trigger plus an amplifier (KJ-3).

07:00
15:00 SGT
MO1550821

> *“we have observed marginal success” … “expand capacity through the utilization of Virtual Machines”*

Microsoft has developed patches and applied them after reboots, with only marginal success, and is now adding virtual machines to the pool. Adding raw capacity is how operators break a self-sustaining overload when no single fix holds: give the system enough headroom that displaced and reconnecting traffic no longer tips the survivors over. It treats the condition, not the cause, and confirms the cause was still unfixed nearly fifteen hours in.

07:29
15:29 SGT
MO1550821

> *“applying an upstream authentication/routing configuration component patch”*

The first fix since 01:45 to report positive results, and the first aimed at the layer in front of the servers rather than the servers themselves. “Upstream” places the component ahead of the “backend servers” of 05:46; “authentication/routing” is what Exchange’s client-access front end does, deciding whose request it is and which mailbox server should receive it. That fits KJ-1 and KJ-2. The 01:45 results did not hold, so treat this as promising until it survives expansion.

Phase 6 09:10 UTC Early stabilisation, and protection for the mail backlog

09:10
17:10 SGT
MO1550821

> *“early signs of stabilization, but service capacity has not yet returned to normal levels”*

The first wording since 01:45 that describes the service as a whole, not a subset of servers. Microsoft does not credit the 07:29 patch: it describes cycling servers through drain, restart, remediation and staged return, the slow way out of a cascade (KJ-2 stays Low until a fix is named).

09:10
17:10 SGT
MO1550821

> *“extending the email timeout window to 48 hours”*

Delayed mail will now be held and retried for two days rather than bounced. That protects messages, but it also means a surge of deliveries when queues drain, possibly while Safe Attachments and Safe Links are still catching up. Users should not resend; SOCs should review what lands during recovery (A-10, A-15).

Read end to end, the updates describe a cascade that outlived its trigger

We assess that Microsoft’s investigation has travelled from the network edge (load, routing), to the health system guarding the server pool, to a process inside each server, back out to a network change, and now to the traffic itself. The evidence fits a self-sustaining overload: a trigger, probably the network change, pushed servers past capacity; health checks removed them; and displaced and reconnecting traffic kept the survivors overloaded after the trigger was reverted. That is why each fix bought partial recovery and none held on its own. At 07:29 the first fix aimed at the routing layer in front of the servers began to show results, and by 09:10 Microsoft saw early stabilisation while it returned servers in stages. If the routing fix proves decisive, the investigation will have ended where it began, with how traffic reaches Exchange. If the PIR confirms this, the lesson is less about one bad change than about a recovery path that could not shed load fast enough.

Was it a cyberattack? Seven of eight attack signatures are absent

If it were an attack, we would expect	What Microsoft and others reported	Reading
A traffic flood at the front door (volumetric DDoS)	At 19:18 Microsoft said total traffic had not risen; load came from servers dropping out.	Not seen
One request type spiking (application-layer flood)	Microsoft examined a rise in one traffic type (18:27), limited traffic by type (23:59), applied “strategic traffic blocking” (04:12) and at 05:46 sought the cause of a “traffic influx”. It frames all of this as stability work.	Open
Signs of intrusion: forensics, credential resets, indicators	None issued. Microsoft has four times said it is not aware of malicious activity. The 07:29 “authentication/routing” patch is a software fix, not a credential reset.	Not seen
A link to an exploited vulnerability	CVE-2026-124194 is client-side: a crafted file opened in Outlook can expose authentication material. Nothing ties it to server availability.	Not seen
Fixes aimed at an adversary: blocking sources, threat hunting	Restarts, health-check changes, traffic shaping and blocking, and two configuration rollbacks, all framed as load management.	Not seen
Failures at other providers at the same time	AWS said it was operating normally. Smaller report spikes for Xbox, GitHub and LinkedIn fit users blaming adjacent services.	Not seen
Officials pointing to an attack	Australia’s Home Affairs Minister said Microsoft told him it did not believe a cyberattack caused the outage.	Not seen
A credible claim of responsibility	Dark Storm Team posted that Outlook was “still offline” about nine hours in, with a check-host link that shows unavailability, not cause. The group has overstated attacks before; we grade the claim E5.	Not seen

The Dark Storm Team claim does not hold up (graded E5)

About nine hours in, the hacktivist group posted on Telegram that Outlook was “still offline”, with a check-host.net link. Check-host shows only that a site is unreachable, which was true for everyone; Check Point notes such “proof links” are the group’s signature, and Graphika found its March 2025 claim against X rested on a report showing one targeted account, not the platform. The post had 93 views; its reach came from aggregators. Constituents should not amplify it.

Analysis of competing hypotheses: the operational explanation is the only one nothing contradicts

ACH tests each hypothesis against every item of evidence and counts what is inconsistent with it; the hypothesis with the least against it survives. C consistent, I inconsistent, N neutral or not diagnostic.

H1 A configuration or software change, then a health-check-driven cascade. **H2** An application-layer attack that exhausts one expensive process without raising total traffic. **H3** A volumetric DDoS. **H4** A malicious change made by an intruder or insider through Microsoft’s own pipeline.

Evidence (UTC)	H1 Operational failure	H2 App-layer attack	H3 Volumetric DDoS	H4 Malicious change
E1 Total traffic did not rise; load came from servers dropping out (19:18)	C	N	I	N
E2 Rise in one type of service traffic (18:27); traffic limited by type (23:59)	C	C	N	N
E3 High CPU in secure-connection setup (19:57, 01:06)	C	C	N	N
E4 Network configuration change correlates with onset; reverted (01:06)	C	N	N	C
E5 Blast radius only Exchange-dependent; Azure and Entra ID unaffected	C	C	N	N
E6 Restored servers fail again under normal traffic (23:59)	C	C	C	N
E7 Fixes all operational; routing-component patch helps (07:29)	C	N	I	N
E8 Microsoft “not aware of any malicious activity” (05:46, 07:00, 07:29, 09:10)	C	I	I	I
E9 Microsoft has named attacks as causes before (Outlook, June 2023; Azure, July 2024) but describes nothing similar here	C	I	I	N
E10 Dark Storm Team claim: after the fact, check-host link only	N	N	N	N
Inconsistent items (fewer is stronger)	**0**	**2**	**4**	**1**

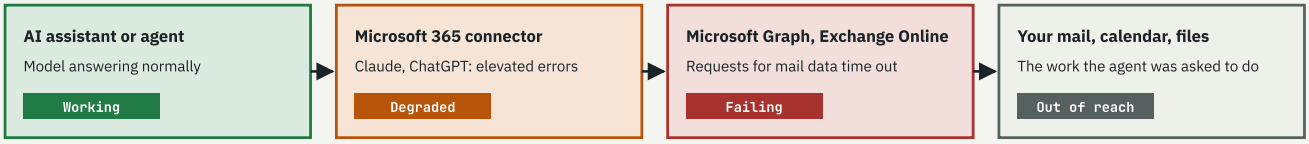
H1 is consistent with every diagnostic item. H3 fails four tests and can be set aside. H2 fails on Microsoft’s statements and silence: it publicly attributed an application-layer attack on Outlook in June 2023 and a DDoS in July 2024, and says it knows of no malicious activity now. The hypothesis least excluded by public evidence is H4: a malicious change would look exactly like the change Microsoft is reverting, and only Microsoft’s own statement argues against it. That is the main reason KJ-4 stays at Moderate confidence, and why question Q-8 asks how the change was made and by whom.

E2 deserves a note. A rise in one request type is what an application-layer attack looks like, and also what a routing change looks like when it shifts which clients reach which servers. Microsoft limited traffic types to find out which one hurt its servers, not to block a source.

Who was hit: beyond the inbox, operators fell back to emergency procedures

Sector or group	What happened	Source
Healthcare	Unity Health Toronto declared a Code Grey (infrastructure failure) because email was down; patient care was not affected. Montreal hospitals reported effects on appointment processing.	CP24/CTV; AAP; CTV Montreal
Transport	VIA Rail services in Canada were affected.	CTV News
Business	Workhorse Group postponed its quarterly results, citing the outage.	Reuters
Education	Universities affected in Canada; Australian remote-learning students could not join Teams classes.	CTV; ABC via Data is CITY
Government	Australia's National Cyber Security Coordinator looked into the outage; Microsoft told the Home Affairs Minister it did not believe it was a cyberattack.	ABC News Australia
AI services	Claude's and ChatGPT's Microsoft 365 connectors failed for mail, calendar and search; Claude for Outlook was affected.	Anthropic and OpenAI status pages
Consumers	Outlook.com, OneDrive, OneNote, To Do, Teams and Microsoft 365 for the web on personal accounts.	Microsoft consumer status
Japan	Downdetector reports rose from about 04:00 JST; Outlook passed 300 by about 06:00 JST.	Nikkei via Data is CITY
Singapore	Affected from 00:17 SGT, peaking near 02:00 SGT; the stalled recovery ran through the working day.	Yahoo News SG; CERT

AI assistants kept working; the work they were connected to did not



The model kept answering; the task still failed. Anthropic opened its connector incident at 17:54 UTC, the same minute Microsoft first posted publicly. Design agents to notice a dead connector, pause, and resume without repeating work (A-9, A-14).

The security controls that sit on mail went dark with it

Microsoft's own list puts Defender for Office 365, Defender XDR, Purview and Intune among the affected services. That is not evidence of an attack, but it does mean mail-borne threats, data-loss controls and device policy may have gone unchecked for the duration. The question after recovery is whether only the consoles failed, or detection and enforcement themselves.

Defender for Office 365, XDR

Microsoft listed
Safe Links and Safe Attachments failures; quarantine views and releases failing; message traces, reporting and Tenant Allow/Block changes delayed; Defender portal pages not loading.

Check after recovery
Detections and alerts for 16:17 UTC onward; allow/block entries actually live; quarantine releases completed (A-3, A-7).

Purview

Microsoft listed
DLP across Microsoft 365, auto-labelling, communication compliance, data classification, audit search and eDiscovery impacted; portal pages not loading.

Check after recovery
Whether DLP policies were enforced on mail sent during the outage; audit-log continuity; eDiscovery holds and exports (A-3).

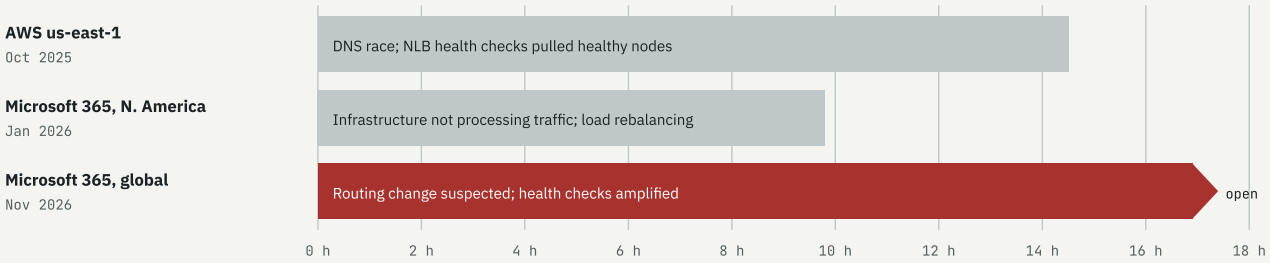
Intune

Microsoft listed
Device enrolment, check-in and compliance policies intermittently impacted.

Check after recovery
Compliance states and policy changes landed; devices that missed check-in re-synced; conditional-access decisions that relied on stale compliance (A-3).

The lesson: health management that can remove the whole pool

Automated health management that ejects failing nodes is a safety system until the failure is shared by every node. Then it becomes the fastest way to remove capacity, and the outage it causes looks like a health-check fault because the checks are what visibly pull the servers. This has happened before, at the largest providers.



Outage	What happened	What changed afterwards
AWS us-east-1, 19–20 Oct 2025	A race condition emptied DynamoDB's DNS record. Network Load Balancer health checks then failed and removed nodes, causing connection errors for about nine hours of a 14.5-hour event.	AWS disabled automatic health-check failover during recovery and added a limit on how much capacity a single NLB can remove after failed checks.
Microsoft 365, 22–23 Jan 2026	Infrastructure in North America stopped processing traffic as expected; recovery took repeated load-balancing changes to clear residual imbalances, nearly 10 hours in all.	Not public in the sources we hold.
Microsoft 365, 12–13 Nov 2026	A suspected routing change, servers failing health checks, survivors overloaded; nearly 17 hours at cut-off.	PIR promised with "defense-in-depth repair items".

Failure indicators to watch on your own platforms: behaviour, not hashes

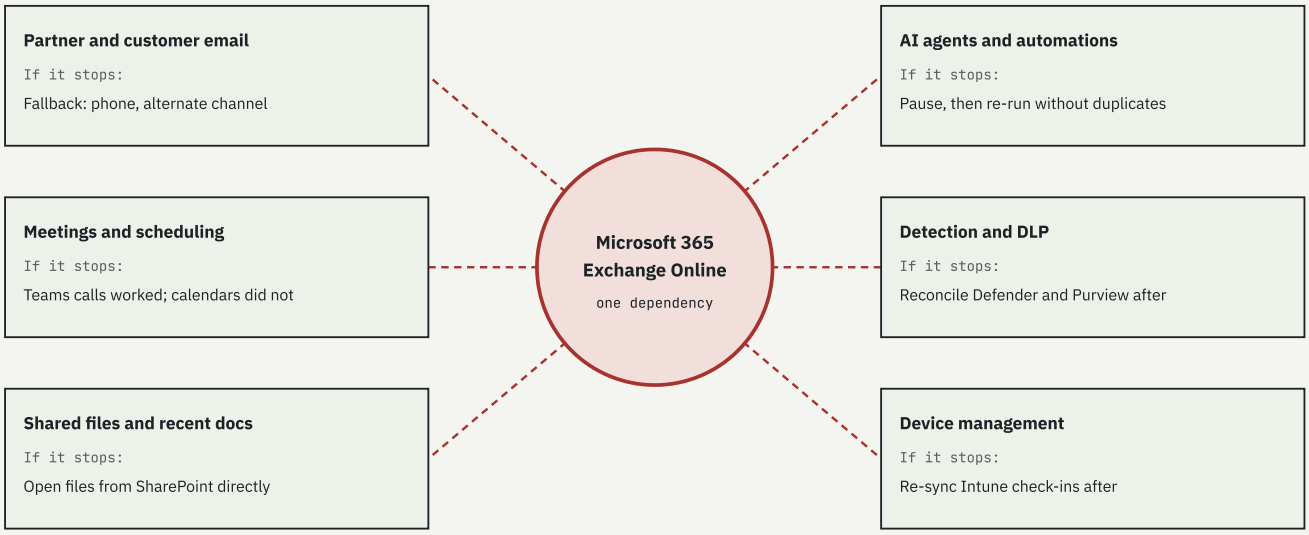
Indicator	What it usually means	Where to watch
Healthy pool size falls while request rate stays flat	Health-driven ejection, not demand. Consider failing open before the pool empties.	Load-balancer target health
Health checks fail on many nodes within minutes	A shared cause, not individual faults. Stop automatic removal and investigate.	Load-balancer health logs
Connection setups or CPU on the termination tier rise without request growth	Connection churn: lost reuse, rerouting or a reconnect storm.	TLS and front-end metrics
Restored nodes fail within minutes of rejoining	Overload persists after the trigger. Shed load and reintroduce traffic gradually.	Node uptime after restore
Traffic mix shifts toward one request type	A routing or client change, or an application-layer attack. Check the source.	Per-protocol request counts
Public outage reports fall while your probes still fail	Attention fading, not recovery. Trust your own telemetry.	Synthetic transactions

Response: what to do today, after recovery, and before the next one

ID	Action	Owner	Horizon	Check
Today				
A-1	Tell users the fault is server-side: no password resets, profile rebuilds or reinstalls. The “authentication” component in Microsoft’s 07:29 update is internal, not user credentials. The same holds for Outlook.com users.	Helpdesks	Now	Script updated
A-2	Track EX1550543 and MO1550821 in the admin centre; use status.cloud.microsoft only if the centre is down.	Tenant admins	Until closed	Both closed
A-4	Warn users about outage-themed phishing and hacktivist claims, such as fake “restore your mailbox” prompts.	CERT comms	7 days	Lure reports triaged
A-6	Safe Links-wrapped URLs fail to open. Tell users not to unwrap or retype them; do not disable Safe Links.	Email admins	Until closed	Advisory issued
A-8	Move urgent communication to phone or another channel. Log work that failed and every setting or workaround changed during the outage.	Business owners; IT	Now	Log kept
After recovery				
A-3	Reconcile Defender for Office 365 alerts, Purview DLP and audit events, and Intune check-ins since 16:17 UTC 12 Nov.	SOC	72 h after fix	Backlogs reconciled
A-7	Re-check Tenant Allow/Block changes and quarantine releases made during the outage.	Email admins	After fix	Changes live
A-9	Re-run failed automations and AI-agent tasks so they do not send the same mail or apply the same update twice.	Automation owners	After fix	No duplicates
A-10	Do not resend mail that is only delayed; Microsoft is holding it for 48 hours. Once service returns, confirm important mail reached its recipients.	Users; business owners	After fix	Critical mail confirmed
A-15	Review mail delivered as queues drain, in Threat Explorer once Defender recovers; Safe Attachments and Safe Links were failing.	SOC	72 h after fix	Backlog reviewed
A-11	Reverse workarounds applied during the outage, such as relaxed policies or temporary forwarding rules.	Tenant admins	After fix	Workarounds removed
Before the next one				
A-5	Route admin-centre service-health alerts to on-call staff; they led Microsoft’s first public post by 94 minutes.	Tenant admins	30 days	Test alert received
A-12	Map which business functions depend on Exchange Online and agree a fallback for each (figure below).	CIO; continuity lead	90 days	Map approved
A-13	Review health-check design on your own platforms: ejection caps, fail-open behaviour, probe isolation.	Platform owners	90 days	Review done
A-14	Test AI agents and automations against a dead connector: they should pause and resume, not retry blindly.	Automation owners	90 days	Test passed

Know your single point of failure before it fails

Mail, meetings, files, detection, device management and now AI agents all rest on the same platform. Drawing that dependency once, with a fallback for each function, turns the next outage from improvisation into a checklist (A-12).



An illustrative map for a typical organisation, not any constituent's actual configuration.

Notice constituents can adapt for their staff

Microsoft 365 (Outlook, Teams calendars and related services) is affected by a global outage on Microsoft's side. Your computer and account are not the problem: please do not reinstall Outlook or change your password.

For anything urgent, use the phone or [alternate channel]. Do not click links in messages asking you to "log in again" or "restore your mailbox".

When service returns, check that important emails you sent today were delivered.

Safe Links failed closed, which pushes users to work around it

Microsoft says users clicking wrapped links cannot reach them, because Safe Links cannot process the URL. That is the safe failure, but it tempts users to unwrap or retype links and tempts administrators to switch the protection off. Both would remove the check precisely while monitoring is degraded. Tell users to wait, and keep the policy on (A-6). Any workaround applied during the outage must be reversed afterwards (A-11).

At 09:10 Microsoft extended its email timeout to 48 hours, so delayed mail should arrive rather than bounce. When queues drain, a surge of messages will land, possibly before Safe Attachments and Safe Links have fully recovered. Tell users not to resend, and review what was delivered during recovery once Defender is back (A-10, A-15).

Intelligence gaps: ten questions Microsoft's post-incident report must answer

ID	Question	Tests
Q-1	Which network setting changed, where was it deployed, and when did its rollout reach each region?	KJ-2
Q-2	Which resource ran out in the secure-connection process: CPU, threads, sockets or memory?	KJ-3
Q-3	Do health probes run through that process or share its resources?	KJ-3
Q-4	Which action restored stability under full traffic: the revert, the rebuild, the blocking, the patches, the added capacity or the 07:29 routing patch?	KJ-2
Q-5	What caused the "traffic influx", and what traffic type was examined at 18:27?	KJ-4
Q-6	Did TM1550488 share a cause, infrastructure or change with this incident?	KJ-2
Q-7	What is the authentication/routing component patched at 07:29, and was it the trigger?	KJ-1, KJ-2
Q-8	Was the network configuration change made through Microsoft's normal pipeline by authorised staff? (ACH H4)	KJ-4
Q-9	Did Defender and Purview miss detections and enforcement, or did only their consoles fail?	A-3
Q-10	Was any delayed mail lost, or delivered without full Defender scanning, as queues drained?	A-15

What is confirmed, what is reported, and what is still our judgement

#	Fact or judgement	Status	Basis (UTC)
1	EX1550543 began at 16:17 and MO1550821 at 18:26, both "service degradation"	Confirmed	Start times in Microsoft's record
2	First admin-centre notice at 16:20; first post on @MSFT365Status at 17:54	Confirmed	Record; X post time decoded from its ID
3	A separate Teams incident (TM1550488) ran 14:45–15:53, blamed on resource contention	Confirmed	Start, end and preliminary cause in the record
4	Load came from servers dropping out, not more traffic; deployments paused	Confirmed	19:06 and 19:18 updates
5	Servers failed health checks; high CPU; removals blocked; alternative check returned a subset	Confirmed	19:43 to 20:56 updates
6	Core process overwhelmed; returned servers failed again under traffic; subset rebuilt	Confirmed	22:00 and 23:59 updates
7	Secure-connection process overwhelmed; a network change that correlates with onset reverted	Confirmed	01:06 update (time per the UPenn mirror)
8	Positive results from restarts, which did not hold	Confirmed	01:45 update; reversed at 03:13
9	Changes "not taking as expected"; "sustained traffic management imbalances"	Confirmed	03:13 X post, time decoded from its ID
10	Traffic protocols adjusted and strategic traffic blocking applied; recovery not sustained	Confirmed	04:12 update
11	Twenty services listed, each failing only where it depends on Exchange	Confirmed	05:46 to 09:10 updates
12	Consumer services hit: Outlook.com, OneDrive, OneNote, To Do, Teams, Microsoft 365 for the web	Confirmed	Consumer status entry, refreshed to 06:30
13	Microsoft seeks the cause of a "traffic influx" and is repairing "backend servers"	Confirmed	05:46 update
14	Microsoft is not aware of any malicious activity; a PIR will follow mitigation	Confirmed	05:46 to 09:10 updates
15	Patches applied after reboots with "marginal success"; virtual-machine capacity being added	Confirmed	07:00 update
16	Patch to an upstream authentication/routing configuration component; positive results on part of the infrastructure	Confirmed	07:29 update
17	Early stabilisation; capacity below normal; email timeout extended to 48 hours	Confirmed	09:10 update
18	Downdetector reports for Outlook peaked at 138,058 near 18:00	Reported	Reuters; reports count complaints, not users
19	Microsoft told Australia's Home Affairs Minister it did not believe it was a cyberattack	Reported	ABC News Australia
20	The failure is confined to Exchange's serving path; shared sign-in is up	Assessed	KJ-1: very likely, High confidence
21	Within Exchange, the fault sits in the client-access front end (CAFE)	Assessed	KJ-1: likely; the 07:29 auth/routing patch fits
22	A change to how traffic is routed into Exchange started the failure	Assessed	KJ-2: likely, Low confidence
23	Health checks amplified the outage but did not cause it alone	Assessed	KJ-3: health-check bug as root cause unlikely, Moderate
24	Not a cyberattack; Dark Storm Team's claim is not credible	Assessed	KJ-4: very unlikely, Moderate; claim graded E5
25	The root cause, the exhausted resource, and which fix worked	Open	Await the PIR (Q-1 to Q-4)
26	What caused the "traffic influx"; the traffic type examined at 18:27	Open	Not explained by Microsoft (Q-5)
27	Any link between TM1550488 and this outage	Open	Microsoft has not said (Q-6)
28	What the 07:29 component is, and whether it was the trigger	Open	Microsoft has not said (Q-7)

References and source grades

Admiralty grades: B2 for Microsoft's record of status and actions, B3 for its cause statements, which it revised during the incident. Press relays C2 unless stated. Data is CITY Media: C2, each claim checked against primary sources. Dark Storm Team's own claim: E5.

Microsoft (primary)

1. Microsoft 365 service health status (consumer and enterprise views); MO1550821 updates of 05:46, about 07:00, 07:29 and 09:10 UTC read directly
   https://status.cloud.microsoft
2. Microsoft incident reports mirror, University of Pennsylvania (EX1550543, MO1550821, TM1550488)
   https://mailservices.isc.upenn.edu/computing/email/penno365/alerts/ms-incidents.html
3. Cloud Admin Hub, Service Communications API mirror: EX1550543
   https://www.cloudadminhub.com/microsoft-365-service-health/ex1550543/
4. Cloud Admin Hub, Service Communications API mirror: MO1550821
   https://www.cloudadminhub.com/microsoft-365-service-health/mo1550821/
5. @MSFT365Status on X; post times decoded from post IDs
   https://x.com/MSFT365Status

Press and analysis

6. Reuters via The Korea Times: Downdetector peak of 138,058
   https://www.koreatimes.co.kr/world/20261113/microsoft-outage-drags-on-for-hours-disrupting-outlook-teams-for-thousands
7. ABC News Australia live blog and report, incl. Home Affairs Minister
   https://www.abc.net.au/news/2026-11-13/microsoft-outage-affects-users-worldwide/107381050
8. CNN, NBC News, The National, ABC7 and XDA Developers live coverage, 12–13 Nov
   https://www.nbcnews.com/tech/tech-news/outlook-widespread-outage-hits-microsoft-apps-rcna609047
9. SecurityWeek on attack evidence and CVE-2026-124194
   https://www.securityweek.com/microsoft-365-outage-disrupts-outlook-exchange-online-and-teams/
10. CP24/CTV: Code Grey at Unity Health Toronto
    https://www.cp24.com/local/toronto/2026/11/12/code-grey-declared-at-toronto-hospital-network-amid-microsoft-outage/
11. CTV Montreal: hospital operations affected
    https://www.ctvnews.ca/montreal/article/microsoft-outage-affecting-chum-appointment-processing-internal-operations/
12. Reuters via The Standard (HK): Workhorse Group results postponed
    https://www.thestandard.com.hk/finance/article/348391/Microsoft-outage-drags-on-for-hours-disrupting-Outlook-Teams-for-thousands
13. Data is CITY Media (Teleport Inc.), independent analysis, 09:10 JST 13 Nov
    https://teleport.jp/dataiscity/news-microsoft-365-outage/
14. Yahoo News Singapore: local timing
    https://sg.news.yahoo.com/

AI providers, threat context and precedents

15. Anthropic status: elevated errors affecting the Microsoft 365 connector
    https://status.claude.com/incidents/sqn3fg1l96dl
16. OpenAI status: elevated errors affecting the Microsoft 365 connector
    https://status.openai.com/incidents/01M78GAT3AK9P9W8Z230KHFVE8
17. Check Point: Dark Storm Team and check-host "proof links"
    https://blog.checkpoint.com/security/dark-storm-team-claims-responsibility-for-cyber-attack-on-x-platform-what-it-means-for-the-future-of-digital-security/
18. Graphika ATLAS: Dark Storm Team overstated claims
    https://public-assets.graphika.com/reports/atlas-highlights-global-hacktivist-threats.pdf
19. AWS: Summary of the Amazon DynamoDB service disruption, us-east-1, Oct 2025
    https://aws.amazon.com/message/101925/
20. heise: AWS root-cause report and NLB capacity-removal limit
    https://heise.de/-10859261
21. The Register: Microsoft 365 outage of 22–23 Jan 2026
    https://www.theregister.com/2026/01/23/microsoft_365_outage/
22. Microsoft Azure status history (30 Jul 2024 post-incident review: DDoS trigger amplified by a defence error)
    https://azure.status.microsoft/en-us/status/history/
23. Help Net Security: Microsoft attributes 30 Jul 2024 outage to DDoS
    https://www.helpnetsecurity.com/2024/07/31/microsoft-azure-ddos/
24. Vumetric summary, incl. June 2023 Anonymous Sudan attacks on Outlook, OneDrive and Azure
    https://cyber.vumetric.com/security-news/2024/07/31/microsoft-says-massive-azure-outage-was-caused-by-ddos-attack
25. Google SRE book: Addressing Cascading Failures
    https://sre.google/sre-book/addressing-cascading-failures/
26. AWS Builders' Library: Implementing health checks
    https://builder.aws.com/content/3Ev53O39izHCtWLzp4XU6t8PC1O/implementing-health-checks